You are building an app that will detect the color scheme of uploaded images.

You are evaluating using the Image Analysis API to detect the dominant background color of an image.

Which color can the API return as a dominant background color?

- azure
- maroon
- silver

This answer is incorrect.

- teal

✓This answer is correct.

Only a certain set of colors can be returned by the API. The set of possible returned colors is black, blue, brown, gray, green, orange, pink, purple, red, teal, white, and yellow.
Color scheme detection - Computer Vision - Azure Cognitive Services | Microsoft Learn
Analyze images - Training | Microsoft Learn

You are developing a custom vision model using Microsoft Azure AI Custom Vision to detect specific objects in images. The dataset includes labeled images for training and validation.

You need to ensure that the trained model meets the required accuracy by assessing its performance.

What action should you take?

- Deploy the model without testing.
- Evaluate precision and recall metrics.

✓This answer is correct.

- Source new images for testing.
- Retrain the model with a new dataset.

Evaluating precision and recall metrics is the most effective way to assess the performance of an object detection model, as these metrics provide a detailed understanding of its accuracy and ability to identify objects correctly. Deploying the model without testing bypasses the critical step of ensuring its reliability and effectiveness. Sourcing new images for testing isn't necessary with a dataset already containing validation images. Retraining the model with a new dataset focuses on improving data quality rather than assessing the current model's performance, making it unsuitable for the given objective.

Test and retrain a Custom Vision model - Training | Microsoft Learn

You are developing a custom vision model using Microsoft Custom Vision to classify images of various plant species. A subset of the dataset has already been labeled, and an initial iteration of the model has been trained.

You need to efficiently improve the model's accuracy by optimizing the labeling process for the remaining untagged images.

Each correct answer presents part of the solution. Which three actions should you perform?

- [ ] Delete all previously tagged images.
- [ ] Manually label all images from scratch.
- [x] Prioritize reviewing tags with high prediction uncertainty.

  ✓This answer is correct.
- [x] Review and confirm suggested tags for each image.

  ✓This answer is correct.
- [ ] Train the model without reviewing tags.
- [x] Use Smart Labeler to suggest tags for untagged images.

  ✓This answer is correct.

To optimize the labeling process and improve the model's accuracy, it is essential to review and confirm suggested tags to ensure data reliability. Prioritizing tags with high prediction uncertainty helps address ambiguous cases effectively, while using the Smart Labeler streamlines the process by leveraging the model's existing capabilities. Deleting previously tagged images would result in the loss of valuable data, training the model without reviewing tags could propagate errors, and manually labeling all images from scratch is inefficient and disregards the benefits of automation.

Use the Azure Document Intelligence Studio - Training | Microsoft Learn

Enrich a search index in Azure AI Search with custom classes and Azure AI Language - Training | Microsoft Learn

You are building an app that will use the Azure AI Custom Vision API to detect when all the spaces in a parking lot are empty.

Which feature of the API should you use?

- image classification
- image description
- object detection

✓This answer is correct.

Object detection is similar to image classification, but it returns the coordinates in an image where the applied label(s) can be found.
Image description analyzes an image and generates a human-readable phrase that describes its contents. Image classification applies one or more labels to an entire image.
What is Custom Vision? - Azure Cognitive Services | Microsoft Learn
Detect objects in images - Training | Microsoft Learn

You are building an app that will use Azure AI Vision to analyze and classify images to build an image library of animals.

You need to configure the classification type for the Azure AI Vision project. The solution must ensure that the images selected only include a single animal.

Which type of classification should you use?

- multiclass

  ✓This answer is correct.
- multilabel
- singleclass
- singlelabel

  This answer is incorrect.
- unilabel

Multilabel classification applies any number of tags to an image (zero or more), while multiclass classification sorts images into single categories (every image you submit will be sorted into the most likely tag). Therefore, using multilabel means that one image can be tagged as both "cat" and "dog" at the same time, although it should be either/or.

Quickstart: Build an image classification model with the Custom Vision portal - Azure Cognitive Services | Microsoft Learn

Classify images - Training | Microsoft Learn

You are building an app that uses Azure AI Video Indexer.

You need to extract keyframes from uploaded video and store them on a disk by using the API.

How should you implement the solution?

- Upload the video and download the ZIP file of the thumbnails.
- Upload the video, download the ZIP file of the thumbnails, and get the thumbnail for each keyframe.

This answer is incorrect.

- Upload the video, get the video index, and get the thumbnail for each keyframe.

✓This answer is correct.

You need to upload the video, get the video index, and get the thumbnail for each keyframe. Three API calls need to be done.
Uploading the video and then downloading the ZIP file of the thumbnails is the path through the Azure portal. You need the index to know the correct parameters for the thumbnail request.
Analyze video - Training | Microsoft Learn

You are building an app that uses the Azure AI Video indexer API to analyze Microsoft Teams meeting recordings. The app will search for images and mentions of competing companies.

Which content model should you use?

- custom brands

✓This answer is correct.

- custom Language
- custom slate

The Custom Brands model supports brand detection from speech and visuals.
Slate detection is used for clapper boards and digital patterns with color bars, and the custom Language model is used to add words that are not in the model.
Customize a Brands model in Azure Video Indexer - Azure - Azure Video Indexer | Microsoft Learn
Analyze video - Training | Microsoft Learn

You are building an app that will extract insights from video files.

You need to identify which service to use. The solution must ensure that you can customize the language model used.

What should you use?

- Azure AI Language
- Azure Communication Services
- Azure AI Vision
- Azure AI Video Indexer

✓This answer is correct.

The only service that can customize the language model for a solution based on gaining insights from videos in Azure AI Video Indexer.

Customize a Language model in Azure Video Indexer - Azure - Azure Video Indexer | Microsoft Learn

Plan and prepare to develop AI solutions on Azure

You are building a mobile app that will enable users to scan street signs and will read out the text on the sign.

You need to recommend a service to use. The solution must minimize development effort.

Which service should you recommend?

- Azure AI Vision

✓This answer is correct.

- Azure AI Custom Vision
- Azure AI Face Service
- Azure AI Document Intelligence

Azure AI Vision is the only service which can achieve the desired result.
Azure AI Custom Vision and Azure AI Face do not offer OCR. Azure AI Document Intelligence is designed for documents, but not images.
OCR - Optical Character Recognition - Azure Cognitive Services | Microsoft Learn
Read Text in Images and Documents with the Computer Vision Service - Training | Microsoft Learn

You are building an Azure AI Language solution.

You need to deploy the solution to a location without internet connectivity.

What should you do?

- (selected) Deploy the solution to a Docker host container.

✓This answer is correct.

- Download the model and deploy it to a virtual machine.
- Use an Azure AI Foundry Service standard instance.

You can use disconnected containers to host Language Understanding on-premises.
There is no virtual machine set up to run a Language Understanding instance and a standard instance still has Language Understanding running in the cloud.
Use Docker containers in disconnected environments - Azure Cognitive Services | Microsoft Learn
Deploy cognitive services in containers - Training | Microsoft Learn

Your company is deploying a generative AI model using Microsoft Azure AI Foundry Service to assist customer support agents by providing suggested responses to user queries.

You need to ensure the model generates outputs that are safe and free from harmful content.

Each correct answer presents part of the solution. Which three actions should you take?

- [x] Conduct red team exercises to test vulnerabilities.

✓This answer is correct.

- [x] Document the model's decision-making logic.

✓This answer is correct.

- [ ] Enable users to provide feedback on responses.
- [x] Integrate Azure AI Foundry Content Safety APIs.

✓This answer is correct.

- [ ] Train the model exclusively on synthetic data.

Conducting red team exercises helps identify vulnerabilities and improve the model's robustness against manipulation. This enhances the security and reliability of the AI system. Documenting the model's decision-making logic provides transparency and supports explainability requirements. This aligns with Responsible AI principles and addresses legal implications. Enabling users to provide feedback on every response may overwhelm users and does not directly address the goal of ensuring content safety. Integrating Azure AI Foundry Content Safety APIs helps identify and filter harmful content in real-time, ensuring the safety and appropriateness of the model's outputs. Training the model exclusively on synthetic data may limit its ability to generate realistic and contextually appropriate responses, reducing its effectiveness.

What is Content Safety - Training | Microsoft Learn

Azure AI Foundry - Training | Microsoft Learn

You are building an app that will identify the core concepts of a document by using Azure AI Foundry Service.

Which endpoint should you use as part of the solution?

- custom Named Entity Recognition (NER)

This answer is incorrect.

- key phrase extraction

✓This answer is correct.

- the Azure AI Vision API

You should use the key phrase extraction endpoint.
The custom NER endpoint will not do key phrase extraction and the Azure AI Vision API can be used to process PDF files but not to extract key phrase detection.
What is key phrase extraction in Azure Cognitive Service for Language? - Azure Cognitive Services | Microsoft Learn
Extract insights from text with the Language service - Training | Microsoft Learn

You plan to build an app that will use Azure AI Foundry Service.

You need to identify the methods that can be used to authenticate to Azure AI Services.

Which two methods can you use? Each correct answer presents a complete solution.

- [ ] a SAML token
- [x] a subscription key

✓This answer is correct.

- [x] Microsoft Entra ID

✓This answer is correct.

- [ ] Kerberos

You can use a single or multi-service subscription keys to authenticate to Azure AI Services. You can also authenticate to Azure AI Services by using a Microsoft Entra ID service principal and role-based access control (RBAC).

Azure AI Services do not support authentication by using SAML tokens or Kerberos.

Authentication in Azure Cognitive Services - Azure Cognitive Services | Microsoft Learn

Secure Cognitive Services - Training | Microsoft Learn

You are building an app that will use Azure AI Custom Vision. The app will be deployed to a virtual machine in Azure.

You enable firewall rules for your Azure AI Services account.

You need to ensure that the app can access the service through a service endpoint.

What should you do?

- Assign a role-based access control (RBAC) role to the Azure AI Custom Vision resource.
- Grant access to a specific virtual network.

  ✓This answer is correct.
- Grant access to an internet IP range.
- Include an access token in the Authorization header.

If you enable the firewall for the Azure AI Services account, you need to allow network access to the service. You can achieve this by either allowing access from a specific virtual network or adding an IP range to the firewall rules. In this situation, the app is deployed to a virtual machine in Azure, which resides in a virtual network. You can provide access to virtual networks in Azure to access specific service endpoints.

Configure Virtual Networks for Azure Cognitive Services - Azure Cognitive Services | Microsoft Learn

Secure Cognitive Services - Training | Microsoft Learn

You are building an app that will use Azure AI Custom Vision. The app will be deployed to an internet host.

You enable firewall rules for your Azure AI Services account.

You need to ensure that the app can access the Azure AI Custom Vision service over the internet.

What should you do?

- Assign a role-based access control (RBAC) role to the Azure AI Custom Vision service.
- Grant access to a specific virtual network.
- Grant access to an internet IP range.

✓This answer is correct.

- Include an access token in the Authorization header.

This answer is incorrect.

If you enable the firewall for the Azure AI Services account, you need to allow network access to the service. You can achieve this by either allowing access from a specific virtual network or adding an IP range to the firewall rules. In this situation, the app is deployed to the internet, and you can provide specific internet hosts access by adding an IP or range of IPs to the firewall rules.

Configure Virtual Networks for Azure Cognitive Services - Azure Cognitive Services | Microsoft Learn

Secure Cognitive Services - Training | Microsoft Learn

You are developing a containerized optical character recognition (OCR)-capable application by using Azure AI Services containers.

While developing the solution, you retrieve a status message of “Mismatch”, and the connection to the AI Services resource fails.

You need to ensure that the solution can connect to the AI Services resource.

What should you do?

- Confirm that the API key is for the correct region.
- Confirm that the API key is for the correct resource type.

  ✓This answer is correct.
- Confirm that the Azure AI Services resource is online.
- Upgrade the Azure AI Services resource to a higher tier.

Mismatch means that the wrong API key has been used. If you have provided an API key or endpoint for a different kind of Azure AI Services resource, you find your API key and service region in the Azure portal, in the Keys and Endpoint section for your Azure AI Services resource.
If the API key is invalid, you must confirm that the API key is for the correct region. If the API key has exceeded the quota, then you can either upgrade your pricing tier or wait for an additional quota to become available. Find your tier in the Azure portal, in the Pricing Tier section of your Azure AI Service resource. The mismatch error would not be generated if the resource was offline.
Cognitive Services containers FAQ - Azure Cognitive Services | Microsoft Learn
Deploy cognitive services in containers - Training | Microsoft Learn

You are building an image moderation solution based on Azure AI Foundry Content Safety.

You migrate images to an Azure Blob Storage location.

You need to configure an access solution to enable the Foundry Content Safety solution to analyze the images.

Which two actions should you perform? Each correct answer presents part of the solution.

- [ ] Assign the Storage Blob Data Contributor/Owner/Reader role to the user-assigned managed identity.
- [x] Assign the Storage Blob Data Contributor/Owner/Reader role to the system-assigned managed identity.

  ✓This answer is correct.
- [x] Enable a system-assigned managed identity for the Content Safety instance.

  ✓This answer is correct.
- [ ] Enable a user-assigned managed identity for the Content Safety instance.

The only way to provide a Content Safety resource with access to images in Azure Blog Storage, is to enable a system-assigned managed identity and to assign the role Storage Blob Data Contributor/Owner/Reader to that managed identity. User-assigned managed identities are not permitted in that scenario.
Quickstart: Analyze image content - Azure AI services | Microsoft Learn

Your organization is implementing Azure AI Foundry Content Safety to moderate user-generated content on a social messaging platform.

You need to configure Foundry Content Safety to identify and restrict harmful text content, including profanities and hate speech, while allowing customization for specific patterns.

Each correct answer presents part of the solution. Which two actions should you take?

- [x] Configure harmful patterns using the Custom Categories API.

✓This answer is correct.

- [x] Enable built-in blocklists in Content Safety Studio.

✓This answer is correct.

- [ ] Enable encryption at rest with customer-managed keys.
- [ ] Use the Analyze Image API for text analysis.

Enabling built-in blocklists in Content Safety Studio is essential as it provides predefined terms to effectively flag harmful content such as profanities and hate speech. Configuring harmful patterns using the Custom Categories API allows for tailored moderation by defining specific patterns that align with organizational needs. Enabling encryption at rest with customer-managed keys enhances data security but does not contribute to the detection or restriction of harmful text content. Using the Analyze Image API focuses on image moderation and is not applicable to the task of moderating text content.

Azure AI services - Training | Microsoft Learn

Use the Azure Document Intelligence Studio - Training | Microsoft Learn

You are deploying an Azure OpenAI in Foundry Models service.

You plan to use your own data in the models you will deploy.

You need to ensure that the model can index your data sources.

Which additional Azure service should you deploy?

- Azure AI Search

✓This answer is correct.

- Content Moderator
- Language
- Personalizer

Azure OpenAI on your data enables developers to use supported AI chat models that can reference specific sources of information to ground the response. Adding this information allows the model to reference both the specific data provided and its pretrained knowledge to provide more effective responses. Azure OpenAI on your data utilizes the search ability of Azure AI Search to add the relevant data chunks to a prompt.

Azure OpenAI on your data still uses a stateless API to connect to the model, which removes the requirement of training a custom model with your data and simplifies the interaction with the AI model. Cognitive Search first finds the useful information to answer the prompt, and Azure OpenAI forms the response based on that information.

Develop a RAG-based solution with your own data using Azure AI Foundry - Training | Microsoft Learn

What are Azure AI services? - Azure AI services | Microsoft Learn

You are building a solution that uses Azure AI Search.

You need to define the field attributes for a field where the search results will include a hit count by category.

Which attribute should you assign to the field?

- `facetable`

✓This answer is correct.

- `filterable`
- `key`

`facetable` is typically used in a presentation of search results that includes a hit count by category.

`Filterable` is referenced in `$filter` queries, and `key` is a unique identifier for documents within the index.

Index overview - Azure Cognitive Search | Microsoft Learn

Create an Azure Cognitive Search solution - Training | Microsoft Learn

You are building a solution that uses Azure AI Search.

You need to save normalized binary files as projections.

Which type of projection should you use?

- files

✓This answer is correct.

- objects
- tables

Tables are used for data that is best represented as rows and columns, or whenever you need granular representations of your data.
Files are used when you need to save normalized, binary image files. Objects are used when you need the full JSON representation of your data and enrichments in one JSON document.
Projection concepts - Azure Cognitive Search | Microsoft Learn
Create an Azure Cognitive Search solution - Training | Microsoft Learn

You are building a solution that uses Azure AI Search.

You need to create a skillset definition.

What are minimum sections you should include in the definition?

- `name`, `description`, and `skills`

✓This answer is correct.

- `name, description, knowledgeStore`, and `encryptionKey`
- `name, description, skills`, and `cognitiveServices`
- `name, description, skills, knowledgeStore`, and `encryptionKey`

Only `name`, `description`, and `skills` are required.

`cognitiveServices` is used for billable skills that call Cognitive Services APIs.

`knowledgeStore` specifies an Azure Storage account and the settings for projecting the skillset output into tables, blobs, and files in Azure Storage.

`encryptionKey` specifies an Azure key vault and customer-managed keys used to encrypt sensitive content (descriptions, connection strings, keys) in a skillset definition.

Create a skillset - Azure Cognitive Search | Microsoft Learn

Create a custom skill for Azure Cognitive Search - Training | Microsoft Learn

You have a web app named App1 that performs customs searches.

You are building a solution that uses Azure AI Search.

You need to include App1 as a custom skill as part of the solution.

Which @odata.type should you use to call App1?

- `Microsoft.Skills.Custom.AmlSkill`
- `Microsoft.Skills.Custom.WebApiSkill`

✓This answer is correct.

- `Microsoft.Skills.Text.CustomEntityLookupSkill`
- `Microsoft.Skills.Util.ConditionalSkill`

`Microsoft.Skills.Custom.WebApiSkill` allows the extensibility of an AI enrichment pipeline by making an HTTP call to a custom web API.

Built-in skills - Azure Cognitive Search | Microsoft Learn

Create a custom skill for Azure Cognitive Search - Training | Microsoft Learn

You are building a knowledge mining solution that will use AI enrichment and Azure AI Search.

You need to create a data structure that will be used to store the enriched and indexed output for downstream apps.

What should you create?

- a knowledge store

  ✓This answer is correct.
- a searchable index
- a searchable store
- an enrichment cache

  This answer is incorrect.

A knowledge store is used for downstream apps, such as knowledge mining and data science. A knowledge store is defined within a skillset. Its definition determines whether your enriched documents are projected as tables or objects (files or blobs) in Azure Storage.

AI enrichment concepts - Azure Cognitive Search | Microsoft Learn

Create a knowledge store with Azure Cognitive Search - Training | Microsoft Learn

Your company processes various forms, including health insurance cards and identity documents, and requires accurate extraction of specific fields.

You need to recommend a solution for extracting structured data from these forms.

What should you do?

- Train a composed model.
- Use Azure OCR.
- Use prebuilt models for Health Insurance Card and ID Document.

✓This answer is correct.

- Use the Layout model.

Prebuilt models for Health Insurance Card and ID Document in Microsoft Azure AI Document Intelligence are the optimal choice for extracting structured data from these forms because they are specifically designed for this purpose, ensuring accuracy and efficiency. Training a composed model introduces unnecessary complexity when prebuilt models are available for these document types. Azure OCR focuses on text extraction without contextual understanding, making it unsuitable for structured data extraction. The Layout model supports key-value pair extraction but lacks the specificity required for extracting specific fields from health insurance cards and identity documents.

Understand AI Document Intelligence - Training | Microsoft Learn

Get started with Azure Document Intelligence

You are building an app that uses Azure AI Services Document Translation.

You need to improve the quality of the translation for user-uploaded documents.

What should you ask the users to include when they upload a document?

- a summary
- the file format
- the source language

✓This answer is correct.

- the writing style

If the language of the content in the source document is known, it is recommended to specify the source language in the request to get a better translation.

Frequently asked questions - Document Translation - Azure Cognitive Services | Microsoft Learn

Translate text with the Translator service - Training | Microsoft Learn

Your organization scans and stores various document types, such as invoices, receipts, and custom forms, in PDF format.

You need to extract structured information from these documents for analysis.

Each correct answer presents part of the solution. Which two actions should you take?

- [ ] Convert the documents into plain text.
- [ ] Train a custom model using labeled datasets in Azure AI Document Intelligence.

✓This answer is correct.

- [ ] Use the General Document model in Azure AI Document Intelligence.
- [x] Use the Layout model in Azure AI Document Intelligence.

This answer is incorrect.

- [x] Use the prebuilt Invoice model in Azure AI Document Intelligence.

✓This answer is correct.

Converting the documents into plain text results in the loss of structural information, which is essential for extracting meaningful data. Training a custom model using labeled datasets in Azure AI Document Intelligence enables accurate extraction of data from unique forms that do not conform to standard templates. The General Document model is deprecated and not recommended for use in new solutions, making it unsuitable for this task. The Layout model can extract text and structure but is not optimized for specific document types like invoices or receipts, which limits its effectiveness for this scenario. The prebuilt Invoice model in Azure AI Document Intelligence is specifically designed to extract key information from invoices, making it an appropriate choice for this task.

Get started with Azure Document Intelligence - Training | Microsoft Learn

Choose a model type - Training | Microsoft Learn

You are creating an assistant based on a generative Azure AI model.

You plan to use the system message component for prompts in the solution.

Which two capabilities does the system message offer for the model? Each correct answer presents part of the solution.

- [ ] defines the data sources that should not be included in the model
- [x] defines what the model should and should not do

  ✓This answer is correct.
- [ ] detects which language is being used in a prompt
- [x] helps define the assistant's personality

  ✓This answer is correct.

The system message is included at the beginning of the prompt and is used to prime the model with context, instructions, or other information relevant to the use case. You can use the system message to describe the assistant's personality, define what the model should and should not answer, and define the format of model responses.

Prompt engineering techniques with Azure OpenAI - Azure OpenAI Service | Microsoft Learn

You are building a web app that will generate images based on user prompts. The app will use the DALL-E 3 Azure OpenAI model.

You need to ensure that HTTP requests against the Azure OpenAI API successfully generate images.

Which three HTTP header properties should you include? Each correct answer presents part of the solution.

- [x] the API version used in this operation

✓This answer is correct.

- [x] the name of the Azure OpenAI service resource

✓This answer is correct.

- [x] the name of the DALL-E 3 model deployment

✓This answer is correct.

- [ ] the quality of the generated images
- [ ] the style of the generated images
- [ ] the user's prompt

The name of the Azure OpenAI resource, the name of the DALL-E 3 model deployment, and the API version to be used are the three required header properties for HTTP requests. The other answers are valid for use in the HTTP body but not the header.

Azure OpenAI Service REST API reference - Azure OpenAI | Microsoft Learn

You are building a web app that will generate images based on user prompts. The app will use the DALL-E 3 Azure OpenAI model.

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Which three HTTP header properties should you include? Each correct answer presents part of the solution.

- [x] the API version used in this operation

✓This answer is correct.

- [x] the name of the Azure OpenAI service resource

✓This answer is correct.

- [x] the name of the DALL-E 3 model deployment

✓This answer is correct.

- [ ] the quality of the generated images
- [ ] the style of the generated images
- [ ] the user’s prompt

The name of the Azure OpenAI resource, the name of the DALL-E 3 model deployment, and the API version to be used are the three required header properties for HTTP requests. The other answers are valid for use in the HTTP body but not the header.

Azure OpenAI Service REST API reference - Azure OpenAI | Microsoft Learn

You are building a web app that will generate images based on user prompts. The app will use the DALL-E 3 Azure OpenAI model.

You need to ensure that HTTP requests against the Azure OpenAI API successfully generate images.

Which HTTP body property should you include?

- the API version
- the Azure OpenAI resource name
- the deployment ID
- the prompt

✓This answer is correct.

The prompt is the only required property to be used in the body of the HTTP request when requesting to generate a new image by using the DALL-E 3 Azure OpenAI API. The other properties are used in the HTTP header.
Azure OpenAI Service REST API reference - Azure OpenAI | Microsoft Learn

You are building a GPT-based chat application that will answer questions about your company.

You plan to use the Using your data feature in Azure OpenAI to ground the model with your company data.

While testing, you discover that some responses are not accurate enough.

You need to configure the Azure OpenAI resource to filter out less-relevant documents for responses.

Which parameter should you configure?

- Content data
- File name
- Retrieved documents
- Strictness

✓This answer is correct.

The Strictness parameter sets the threshold to categorize documents as relevant to your queries. Raising the Strictness parameter value means a higher threshold for relevance and filters out more less-relevant documents for responses. Retrieved documents specifies the number of top-scoring documents from your data index used to generate responses. Content data specifies the fields in your index that contain the main text content of each document. File name specifies the field in your index that contains the original file name of each document.
Using your data with Azure OpenAI Service - Azure OpenAI | Microsoft Learn

You are building a GPT-based chat application that will answer questions about your company.

You plan to test the application by using strategies defined by Microsoft best practices.

Which three prompt engineering strategies should you consider while testing the application? Each correct answer presents a complete solution.

- [x] Be Descriptive

✓This answer is correct.

- [ ] Be minimalistic
- [ ] Be simple
- [x] Be Specific

✓This answer is correct.

- [x] Order Matters

✓This answer is correct.

Be Specific means to leave as little to interpretation as possible. Be Descriptive means to use analogies. Order Matters means that the order in which you present information to the model can affect the output. Therefore, those three are valid best practices. Be simple and be minimalistic do not produce the best results and are, therefore, not best practices

Azure OpenAI Service - Azure OpenAI | Microsoft Learn

You are creating an application that will use Azure OpenAI REST API services. The application uses a REST call to a DALL-E model to generate images. The three parameters in the REST call are `prompt`, `n`, and `size`.

What does the `size` parameter indicate?

- the number of responses that you want returned
- the size of the images in bytes
- the size of the images in kilobytes
- the size of the images in pixel resolution

✓This answer is correct.

To make a REST call to the services, you need the endpoint and authorization key for the Azure OpenAI service resource you provisioned in Azure. You initiate the image generation process by submitting a `POST` request to the service endpoint that has the authorization key in the header. The request must contain the following parameters in a JSON body:

`prompt`: The description of the image to be generated

`n`: The number of images to be generated

`size`: The resolution of the image to be generated (*256x256*, *512x512*, or *1024x1024*)

Use the Azure OpenAI REST API to consume DALL-E models - Training | Microsoft Learn

An organization plans to use Azure OpenAI in Foundry Models to process user input and generate content.

You need to create visual content from text descriptions.

What should you use?

- Azure AI Search
- DALL-E

✓This answer is correct.

- GPT-4
- Text Embedding Models

DALL-E is the appropriate choice for creating visual content from text descriptions, as it is specifically designed for this purpose. Azure Cognitive Search focuses on information retrieval and does not support visual content creation. GPT-4 is optimized for text-based tasks and lacks the capability to generate visual content. Text Embedding Models are intended for embedding tasks and are unrelated to the goal of creating visual content from text descriptions.

Plan and prepare to develop AI solutions on Azure - Training | Microsoft Learn

Developer tools and SDKs - Training | Microsoft Learn

Your company uses an Azure OpenAI in Foundry Models service to generate images based on text prompts.

You need to configure the API request to ensure optimized image quality and resolution.

Each correct answer presents part of the solution. Which two actions should you take?

- [ ] Set 'n' to 10.
- [ ] Set 'output_format' to JPEG.
- [x] Set 'quality' to hd.

✓This answer is correct.

- [x] Specify 'size' as 1024x1792.

✓This answer is correct.

To optimize image quality and resolution, 'quality' should be set to hd to ensure finer details and better consistency, and 'size' should be specified as 1024x1792 to meet the required resolution. Setting 'n' only determines the number of images generated and does not impact quality or resolution. Configuring 'output_format' affects the image format but does not influence image quality or resolution.

How to use Azure OpenAI image generation models - Training | Microsoft Learn

You are deploying a generative AI solution using an Azure OpenAI in Foundry Models service to process customer feedback and generate summaries.

You need to optimize the solution for performance and ensure its responses align with the organization's tone and style.

Each correct answer presents part of the solution. Which three actions should you perform?

- [ ] Deploy the model on edge devices.
- [x] Enable support for multiple languages.

  This answer is incorrect.
- [x] Fine-tune the model with customer feedback data.

  ✓This answer is correct.
- [ ] Set up monitoring in Azure to track response accuracy and resource usage.

  ✓This answer is correct.
- [x] Use prompt engineering to refine the model's output.

  ✓This answer is correct.

To optimize the generative AI solution for performance and ensure its responses align with the organization's tone and style, it is essential to set up monitoring in Azure to continuously evaluate the model's performance and resource usage. Fine-tuning the model with customer feedback data customizes the solution to meet specific organizational requirements, ensuring alignment with tone and style. Additionally, using prompt engineering refines the model's responses by structuring input effectively, enhancing the quality and relevance of the generated content. Deploying the model on edge devices and enabling support for multiple languages do not directly address the stated goals, as they focus on deployment strategy and additional features rather than performance optimization or alignment with tone and style.

Fine-tune a language model with Azure AI Foundry - Training | Microsoft Learn

Your team is deploying an AI agent to perform actions based on user requests, such as scheduling meetings and sending notifications.

You need to integrate the AI agent with tools that enable it to perform these actions programmatically.

What should you do?

- Add custom functions.

✓This answer is correct.

- Configure static templates.
- Enable default model capabilities.
- Use a pre-trained chatbot framework.

Adding custom functions enables the AI agent to dynamically execute specific actions such as scheduling meetings and sending notifications, making it the most suitable choice for the scenario. Configuring static templates does not provide the flexibility needed for dynamic action execution, limiting its effectiveness. Default model capabilities lack the ability to perform specific programmatic actions, making them insufficient for the task. Using a pre-trained chatbot framework without customization fails to address the specific requirements for executing actions programmatically, reducing its applicability in this context.

Integrate custom tools into your agent - Training | Microsoft Learn

Your organization is developing an AI agent to assist with document retrieval and analysis using Microsoft Azure services.

You need to enable the AI agent to access external data sources for processing documents.

What should you use?

- ( ) Configure a custom model in Azure Machine Learning.
- ( ) Enable default capabilities in Azure Document Intelligence.
- ( ) Use Azure AI Vision Service.
- (•) Use Azure AI Search with built-in skills.

✓This answer is correct.

Azure AI Search is the most suitable solution because it provides the required functionality for accessing and analyzing external documents with built-in skills, which aligns with the scenario's needs. Configuring a custom model in Azure Machine Learning focuses on document analysis but does not address the retrieval aspect, making it unsuitable. Enabling default capabilities in Azure Document Intelligence lacks the specific retrieval functionality needed for the task. Azure AI Vision Service can perform OCR but is not well suited for document processing and lacks the retrieval aspect.

Developer tools and SDKs - Training | Microsoft Learn

Your organization plans to develop an AI solution using Microsoft Azure AI Foundry Agent Service to automate customer support workflows.

You need to configure the agent to interact with external data sources and execute actions based on user queries.

Each correct answer presents part of the solution. Which two actions should you perform?

- [ ] Add Bing search tools in the agent configuration.

✓This answer is correct.

- [x] Configure Azure Functions in the agent configuration file.

✓This answer is correct.

- [ ] Deploy a containerized image.
- [x] Set up a SQL database.

This answer is incorrect.

- [ ] Use Computer Vision.

Adding Bing search tools enables the agent to retrieve relevant information from external sources, enhancing its ability to respond accurately to user queries. Configuring Azure Functions allows the agent to execute specific actions programmatically, which is crucial for automating workflows based on user inputs. Deploying a containerized image is not necessary to configure the agent's capabilities. Setting up a SQL database is not required for configuring the agent's interaction capabilities, as it does not directly contribute to the goal of automating customer support workflows. Using Computer Vision is unrelated to the task of configuring the agent for customer support workflows, as it focuses on image analysis rather than interaction with external data sources.

Develop an AI agent with Azure AI Foundry Agent Service - Training | Microsoft Learn

Your organization is using Azure AI Foundry Agent Service to automate customer support workflows.

You need to configure an agent to interact with external APIs and use real-time data for responses.

Each correct answer presents part of the solution. Which three actions should you perform?

- [x] Define tools for API access.

✓This answer is correct.

- [ ] Deploy the agent using Azure SDK.

✓This answer is correct.

- [ ] Manually parse API responses in code.
- [x] Configure the Bing Search tool.

✓This answer is correct.

- [x] Set up an Azure AI Search resource.

This answer is incorrect.

- [ ] Use Blob Storage for conversation state.

Defining tools for API access ensures the agent can interact with external APIs effectively. Deploying the agent using Azure SDK makes it operational within the Azure environment, enabling API interactions. Configuring the Bing Search tool enables the agent to access real-time data. Setting up an Azure AI Search resource is optional and does not directly contribute to enabling API interaction for the agent. Manually parsing API responses is unnecessary because Azure AI Foundry Agent Service automates tool calling and response handling. Using Blob Storage for conversation state is unnecessary because Azure AI Foundry Agent Service manages conversation state through threads.

Develop an AI agent with Azure AI Foundry Agent Service - Training | Microsoft Learn

You need to build an app that will summarize text documents into key phrases.

Which Azure AI Language feature should you recommend?

- Language Summarization
- key phrase extraction

✓This answer is correct.

- Named Entity Recognition (NER)
- Personally Identifiable Information (PII) detection

Key phrase extraction is used to quickly identify the main concepts in text, whereas the other features do not return key phrases from longer text documents.

What is key phrase extraction in Azure Cognitive Service for Language? - Azure Cognitive Services | Microsoft Learn

Extract insights from text with the Language service - Training | Microsoft Learn

You are building an app that will analyze meeting recordings and identify who is speaking at which moment in time.

You need to configure a voice profile for the app.

Which type of voice profile should you use?

- Speaker identification

✓This answer is correct.

- text-dependent verification
- text-independent verification

Text-independent verification means that speakers can speak in everyday language in enrollment and verification phases.
Text-dependent verification means that speakers need to choose the same passphrase to use during both enrollment and verification phases. This is the voice profile that should be used when configuring a voice profile for the app.
Speaker identification helps you determine an unknown speaker's identity within a group of enrolled speakers.
Speaker recognition overview - Speech service - Azure Cognitive Services | Microsoft Learn
Create speech-enabled apps with the Speech service - Training | Microsoft Learn

You are building an app that will enable users to create notes by using speech.

You need to recommend the Azure AI Speech service model to use. The solution must support noisy environments.

Which model should you recommend?

- base
- base with customizations
- custom speech-to-text

✓This answer is correct.

- default

This answer is incorrect.

The custom speech-to-text model is correct, as you need to adapt the model because a factory floor might have ambient noise, which the model should be trained on.

Speech-to-text FAQ - Azure Cognitive Services | Microsoft Learn

Create speech-enabled apps with the Speech service - Training | Microsoft Learn

You are building a custom translation model.

You need to evaluate the precision of the text that you translated by using a Bilingual Evaluation Understudy (BLEU) score.

Which scale is used for the score?

- between 0 and 1
- between 0 and 100

✓This answer is correct.

- low, medium, and high

A BLEU score is a number between zero and 100. A score of zero indicates a low-quality translation, where nothing in the translation matches the reference. A score of 100 indicates a perfect translation that is identical to the reference. It is unnecessary to attain a score of 100. A BLEU score between 40 and 60 indicates a high-quality translation.

Custom Translator for beginners - Azure Cognitive Services | Microsoft Learn

Translate text with the Translator service - Training | Microsoft Learn

You are creating an orchestration workflow for Language Understanding.

You need to configure workflows for multiple languages. The solution must minimize administrative effort.

What should you create for each language?

- a new deployment
- separate models
- separate training jobs
- separate workflow projects

✓This answer is correct.

Orchestration workflow projects do not support the multilingual option, so you need to create a separate workflow project for each language.

Language support for orchestration workflow - Azure Cognitive Services | Microsoft Learn

Build a Language Understanding model - Training | Microsoft Learn

You are configuring a question answering solution.

You execute the following API call and receive the following error.

```
"synonyms": [
        {
              "alterations": [
                    "fix problems",
                    "troubleshoot",
                    "#diagnostic",
                    ]
        },
...
```

You need to ensure that the API call executes successfully.

What should you do?

- Modify the order of the synonyms.
- Remove any question and answer pairs from the call.
- Remove any special characters from the call.

✓This answer is correct.

Special characters are not allowed for synonyms.
Synonyms can be added in any order, and the ordering is not considered in any computational logic. Synonyms can only be added to a project that has at least one question and answer pair.
Improve the quality of responses with synonyms - Azure Cognitive Services | Microsoft Learn
Build a question answering solution - Training | Microsoft Learn

You plan to build a chatbot that will help users answer FAQs.

You need to identify which scenarios are suitable for use with the Azure AI Language question answering service.

Which three scenarios should you identify? Each correct answer presents a complete solution.

- [ ] when you have a bot conversation that includes dynamic information
- [x] when you have a bot conversation that includes static information

  ✓This answer is correct.
- [ ] when you have dynamic information in a knowledge base of answers
- [x] when you have static information in a knowledge base of answers

  ✓This answer is correct.
- [x] when you need to provide the same answer to a request, question, or command

  ✓This answer is correct.
- [ ] when you need to provide unique answers to each request, question, or command

Question answering only works with static information, not with dynamic information. In addition, it will always provide the same answer to the same question.
What is question answering? - Azure Cognitive Services | Microsoft Learn
Build a question answering solution - Training | Microsoft Learn

You are building a chatbot that will use the Azure AI Language question answering service.

You need to identify the operational costs for the service.

Which two parameters will influence the costs? Each correct answer presents a complete solution.

- [ ] the number of assigned metadata tags
- [ ] the number of knowledge base editors
- [ ] the number of supported languages
- [x] the required throughput

✓This answer is correct.

- [x] the size and the number of knowledge bases

✓This answer is correct.

The throughput, the size, and the number of knowledge bases affect the pricing tier, whereas the other parameters do not affect pricing.

Azure resources – question answering – Azure Cognitive Services | Microsoft Learn

Build a question answering solution – Training | Microsoft Learn